

MOLECULAR CHARACTERIZATION OF GANGLIOSIDE-PEPTIDE INTERACTIONS

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ABSTRACT

Gangliosides are glycosphingolipids found in small quantities in cell membranes and are characterized by remarkable molecular diversity. Recently, a renewed interest in the analysis of these species¹ has reinforced our conviction that a detailed and more thorough characterization of these molecules is necessary to fully understand their biological functions at the molecular level. To address this challenge, we developed an innovative mass spectrometry-based workflow for the precise and quantitative analysis of gangliosides. This approach combines MALDI-TOF mass spectrometry and ion mobility spectrometry, enabling detailed characterization of gangliosides at multiple structural levels, from ganglioside classes to individual molecular species and their potential isomers. The method was first established using a complex commercial gangliosides mixture before being applied to cellular extracts.

Gangliosides are key regulators of homeoprotein paracrine signaling. In order to better understand their precise role, we investigate their molecular interaction with En2H₃, a cell-penetrating peptide (CPP) derived from the homeoprotein Engrailed2, using complementary biophysical and cellular approaches. *In vitro* studies², including NMR spectroscopy, differential scanning calorimetry (DSC), and affinity photo-cross-linking, are performed on membrane models (MLVs) containing the well-characterized commercial gangliosides mixture. In parallel, cellular assays quantify En2H₃ internalization in cells expressing distinct ganglioside profiles, which will be characterized using our newly developed analytical workflow.

REFERENCES

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